

Minutes of The Sixth Meeting

OS1 group B

10th Apr 2025

Chair Orvila Sarker

Secretary Yifan Gu

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Apologies None.

1. Time and Place

The fifth team meeting for the MCI OS1 project was held via **Zoom video conference** from **4:00 PM to 4:35 PM** on **3rd April**.

2. Quorum Announcement

The Chairman announced that a quorum of the group was present, and that the meeting, having been duly convened, was ready to proceed with its business.

3. Summary of Previous Meeting

The team finalized the business report, delegated project tasks, and initiated work within their respective areas of responsibility.

4. Group Milestone

4.1. Business Report

The team adjusted and modified all aspects of the project according to the supervisor's feedback, and the results were remarkable.

4.2. Backend code framework construction

To support an Augmented Software Security Knowledge Base, the backend was implemented using FastAPI, following a modular design pattern under the app/pipeline directory. The system integrates multiple components including classification, sentiment analysis, and retrieval-augmented generation (RAG).

4.3. Feature Implementation and Prototype Integration

- Similarity has been integrated into the backend code framework. In the current system, the processing of "similarity" adopts a dual mechanism of vector semantic retrieval and LLM semantic filtering. The question entered by the user is first embedded through the language model to build a vector table. The system will generate a structured answer based on the retrieved documents, and finally only retain the answers that are judged to be "relevant" for display.
- Sentiment analysis has been integrated into the backend code framework. The library currently used is VaderSentimentAnalyzer, a sentiment analysis component based on VADER, encapsulated as a Runnable type of LangChain. It judges the input text as positive, neutral or negative based on the composite sentiment score.
- The classifier has not yet been integrated into the backend code framework. The idea of the classifier is to train an SVM model with raw data, and call the trained model in the project to classify user input questions. It has been encapsulated into a class, which is convenient for integration into the backend code in the next step.

5. Submission

Submit the final business report. The project code is updated on Github.

6. Next Steps

6.1. Integrate the classifier into the code framework

It has been encapsulated into the class and will be integrated into the overall code framework in the next step.

6.2. Similarity Improvement

Instead of giving users a similarity value, replace it with an appropriate and explanatory text description.

6.3. Sentiment analysis improvement

Use LIWC to perform sentiment analysis on questions asked by users.

7. Adjournment

The next team meeting is tentatively scheduled for **Week 7 after the mid-semester break**, on **May 1st at 3:30 PM**.